Executive Summary – Customer Churn Analysis

This project analyzes customer churn behavior using demographic, service usage, and billing data. The objective is to identify key drivers of churn and provide actionable insights to improve customer retention.

The overall churn rate is approximately **26.5%**, meaning that **1 out of every 4 customers (≈ 27%) leaves the service**, which is a critical business concern impacting revenue and growth.

Customer Tenure Insights

Customer tenure shows a strong inverse relationship with churn:

- Customers with **0–2 months tenure** contribute to **over 50% of total churn cases**
- Customers staying beyond **12 months have significantly lower churn (<15%)**

This indicates that **early-stage customer experience is the most crucial phase**, and improving onboarding can reduce churn significantly.

Contract Type Analysis

Contract duration is one of the strongest predictors of churn:

- **Month-to-month contracts: ~42–45% churn rate**
- **1-year contracts: ~11–12% churn rate**
- **2-year contracts: ~2–3% churn rate**

This clearly shows that customers on flexible plans are **~15x more likely to churn** compared to long-term contract users.

Service Usage Patterns

Value-added services play a major role in retention:

- Customers **without Online Security** → **~40% churn**
- Customers **without Tech Support** → **~38% churn**
- Customers **without Device Protection** → **~35% churn**
- Customers using these services show churn rates reduced to **15–20%**

This highlights that **bundled services improve customer stickiness by nearly 50%**.

Internet Service Type

- **Fiber optic users: ~40–42% churn rate**
- **DSL users: ~18–20% churn rate**
- **No internet service:** very low churn

Despite being a premium service, fiber users churn nearly **2x more**, suggesting possible issues with **pricing, reliability, or expectations mismatch**.

Demographic Factors

- **Senior citizens: ~41–42% churn rate**
- **Non-senior customers: ~23–24% churn rate**
- **Gender:** No significant difference (≈ equal churn distribution)

Senior citizens are **almost 2x more likely to churn**, indicating a need for better support or tailored plans.

Payment Method & Billing

Billing and payment methods significantly impact churn:

- **Electronic check users: ~45% churn rate** (highest)
- **Mailed check / bank transfer / credit card: ~15–20% churn**

Customers using electronic checks are **more than 2x likely to churn**, indicating possible friction in payment experience.

Monthly Charges Impact

- Customers with **higher monthly charges (>₹70 equivalent)** show **higher churn (~35–40%)**
- Lower-paying customers churn less (~15–20%)

This suggests **price sensitivity and perceived value issues**.

Key Business Insights & Recommendations

- 🔴 **Early churn (≈50%) happens in first 2 months**
  → Improve onboarding, offer welcome benefits, and proactive engagement
- 🔴 **Month-to-month customers (≈45% churn)**
  → Incentivize long-term contracts with discounts or loyalty benefits
- 🔴 **Lack of add-on services increases churn by ~2x**
  → Bundle services like security & tech support
- 🔴 **Fiber optic users (≈40% churn)**
  → Investigate service quality, pricing, and customer expectations
- 🔴 **Electronic check users (≈45% churn)**
  → Simplify payment process and promote auto-pay options
- 🔴 **Senior citizens (≈42% churn)**
  → Provide personalized support and easy-to-use services

Conclusion

The analysis clearly shows that churn is driven by a combination of **contract flexibility, service engagement, pricing, and customer experience**. By targeting high-risk segments (new users, month-

to-month customers, and low-engagement users), the company can potentially reduce churn by **15–20%**, leading to significant revenue retention.